



UNIT 1: Machine Learning Foundation (Lecture 1)

The Story of a Spam Filter



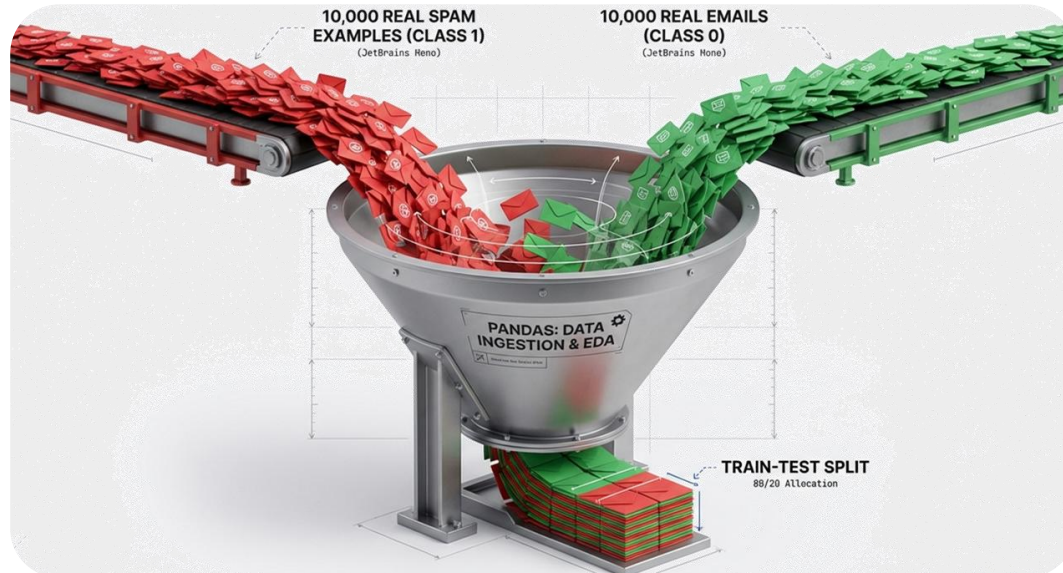
The Story of a Spam Filter

- Imagine you've been hired by a major email provider to block "spam" emails.
- Initially, you write several of rules, like:
 - "If the subject contains 'WINNER', it's spam"
 - "If the sender is unknown and asks for money, it's spam"
 - ...
- **But the spammers are clever.**
- They start writing "W-I-N-N-E-R" or "Congratulations!"
- 🤔 Your rules fail. You're exhausted.
- Then, You realize **you can't manually code every possibility.**



The Story of a Spam Filter

- You realize you can't manually code every possibility.
 - Instead, you show the computer 10,000 examples of real spam and 10,000 real emails.
 - The **computer notices (or, learns)** that spam emails often have certain patterns of words you hadn't even thought of.



- Suddenly, the system starts blocking spam it has never seen before. That is the **heart of Machine Learning.** 💖

Pattern recognition beyond the training set.

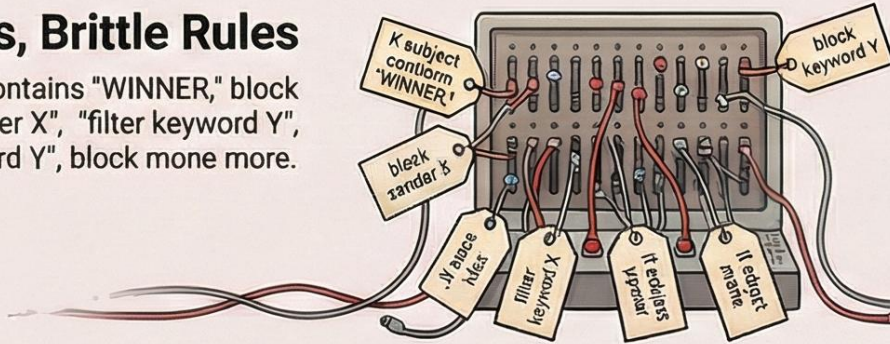


How AI Beats Spam: From Manual Rules to Machine Learning

THE OLD WAY: MANUAL FAILURE

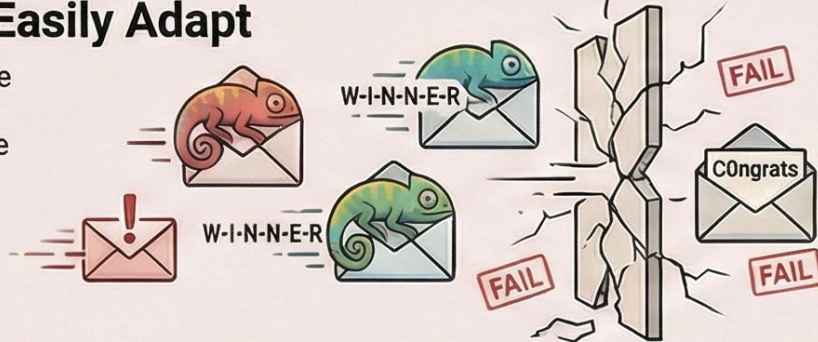
Endless, Brittle Rules

if subject contains "WINNER," block
"block sender X", "filter keyword Y",
filter keyword Y", block none more.



Spammers Easily Adapt

Clever variations like "W-I-N-N-E-R" or "C0ngrats" break the manual rules.



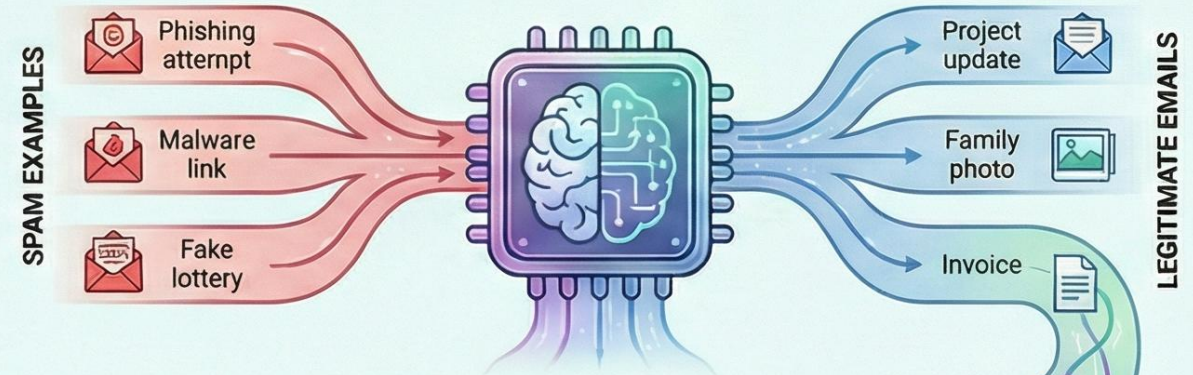
An Impossible Task

Manually coding for every possible spam variation is unsustainable and ineffective.

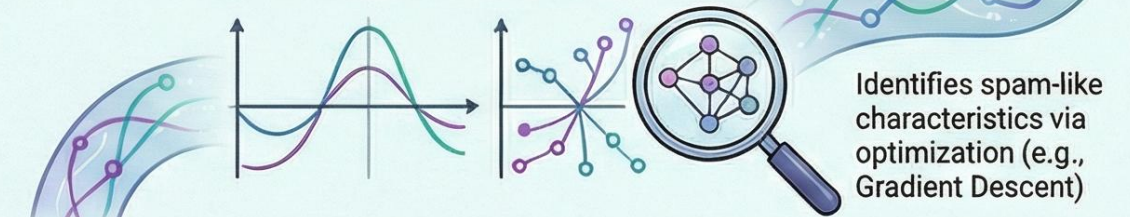


THE ML WAY: LEARNING FROM DATA

1. Feed the Machine Data



2. The Machine Finds Patterns

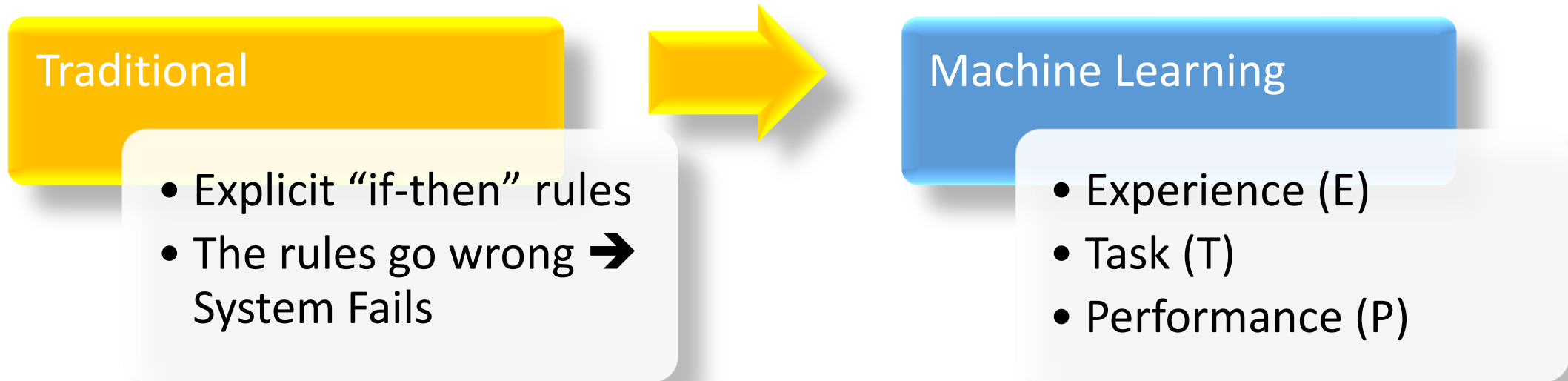


3. Generalize to Block New Threats



The Story of a Spam Filter

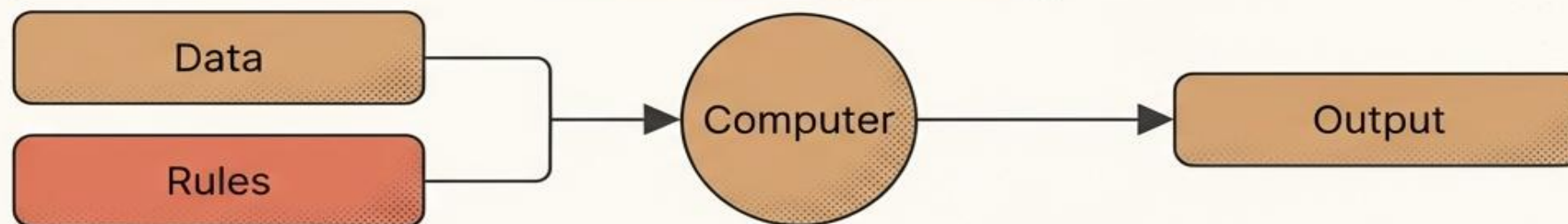
- **Traditional Programming:**
 - You write explicit "if-then" rules.
 - If the rules are wrong, the system fails.
- **Machine Learning:**
 - We provide the system with **Experience (E)** (past emails) for a **Task (T)** (identifying spam) to improve its **Performance (P)** (accuracy).



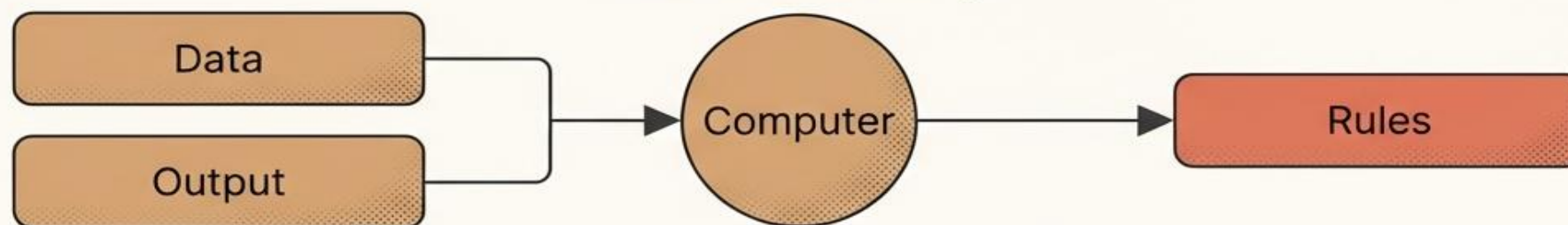
The Paradigm Shift

From Logic to Learning

Traditional Programming



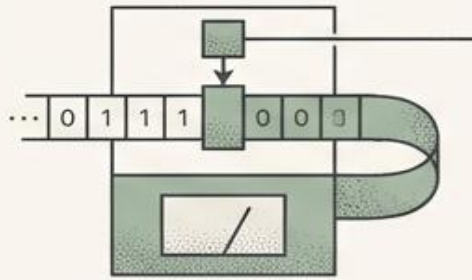
Machine Learning



In traditional programming, humans hard-code the rules. In Machine Learning, the system infers the rules by analyzing the relationship between data and answers.

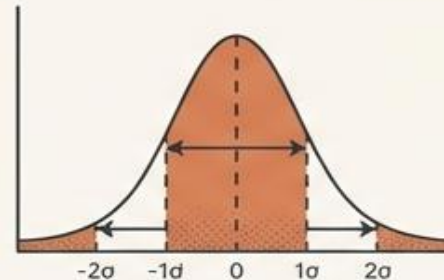
Introduction to Machine Learning

A Brief History of Intelligence



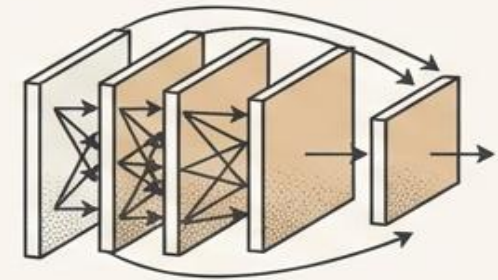
1950s - The Dawn

Turing Test & Perceptrons
The fundamental question:
"Can machines think?"



1980s - The Probabilistic Turn

Rise of Statistical Learning
Moving from rigid logic to probability-based models.



2010s - The Deep Learning Boom

Big Data + GPUs
Massive datasets (ImageNet) enable deep neural networks.

Introduction to Machine Learning

Defining Machine Learning

1959

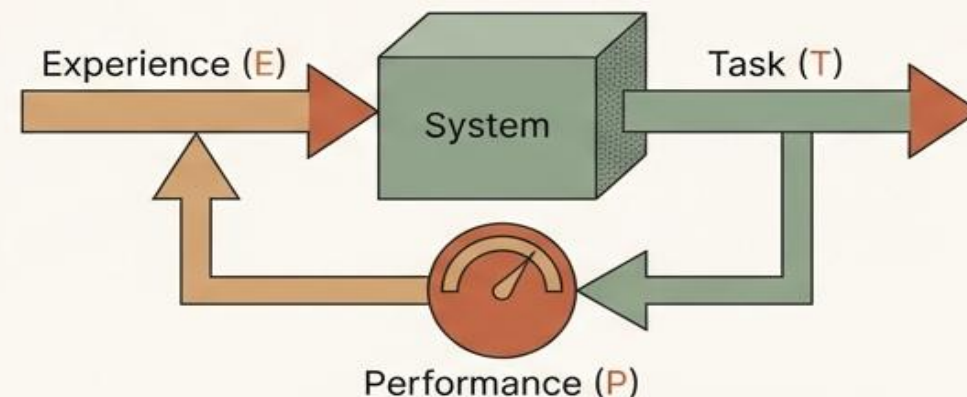
“Field of study that gives computers the ability to learn without being explicitly programmed.”

— Arthur Samuel
Inter

1997

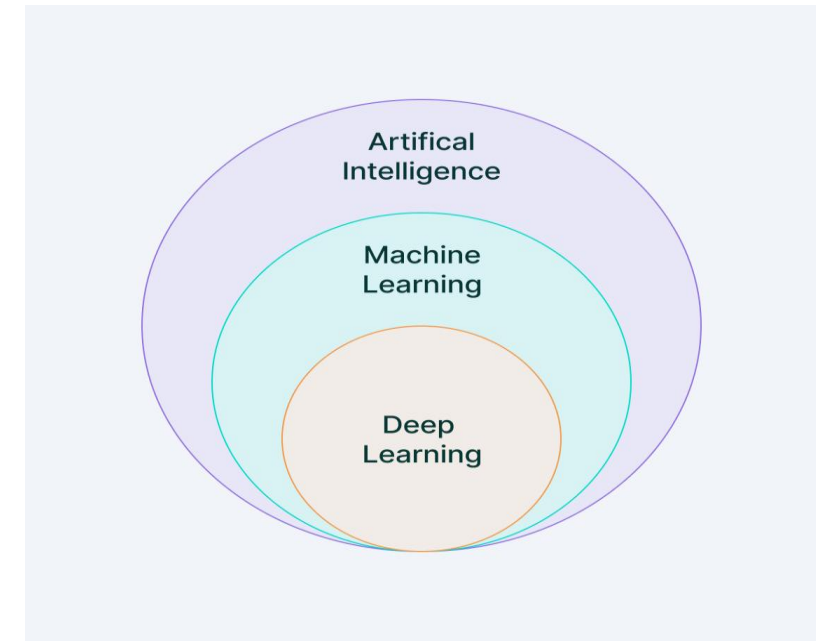
A computer program is said to learn from Experience (E) with respect to some class of Tasks (T) and Performance measure (P), if its performance at tasks in T, as measured by P, improves with experience E.

— Tom Mitchell



What is Machine Learning?

- Machine learning is a branch of [artificial intelligence \(AI\)](#).
- Enables machines to learn from data.
- Improves performance without explicit programming.
- Based on algorithms and statistical models.
- Focuses on pattern recognition and predictions.
- Used in diverse fields like healthcare, finance.



MACHINE LEARNING

> DEFINITION

A subset of AI that enables systems to automatically learn and improve from experience without being explicitly programmed

“A computer program is said to learn from **experience E** with respect to some class of **tasks T** and performance **measure P**, if its performance at tasks in T, as measured by P, improves with experience E.”



What is AI?

Artificial Intelligence (AI): Computer systems designed to perform complex tasks.

Capable of tasks that historically required human intelligence.

Examples of tasks:

- Reasoning – drawing conclusions from information.
- Decision-making – choosing optimal actions.
- Problem-solving – finding solutions to complex challenges.



Need of Machine Learning:

- The need for machine learning arises from various challenges and opportunities in different domains. Here are some key reasons why machine learning is important:
- Handling Complex and Large Data Sets
- Automation and Efficiency
- Pattern Recognition
- Personalization
- Prediction and Forecasting



- Natural Language Processing (NLP)
- Fraud Detection and Security
- Healthcare and Medicine
- Autonomous Systems
- Continuous Improvement

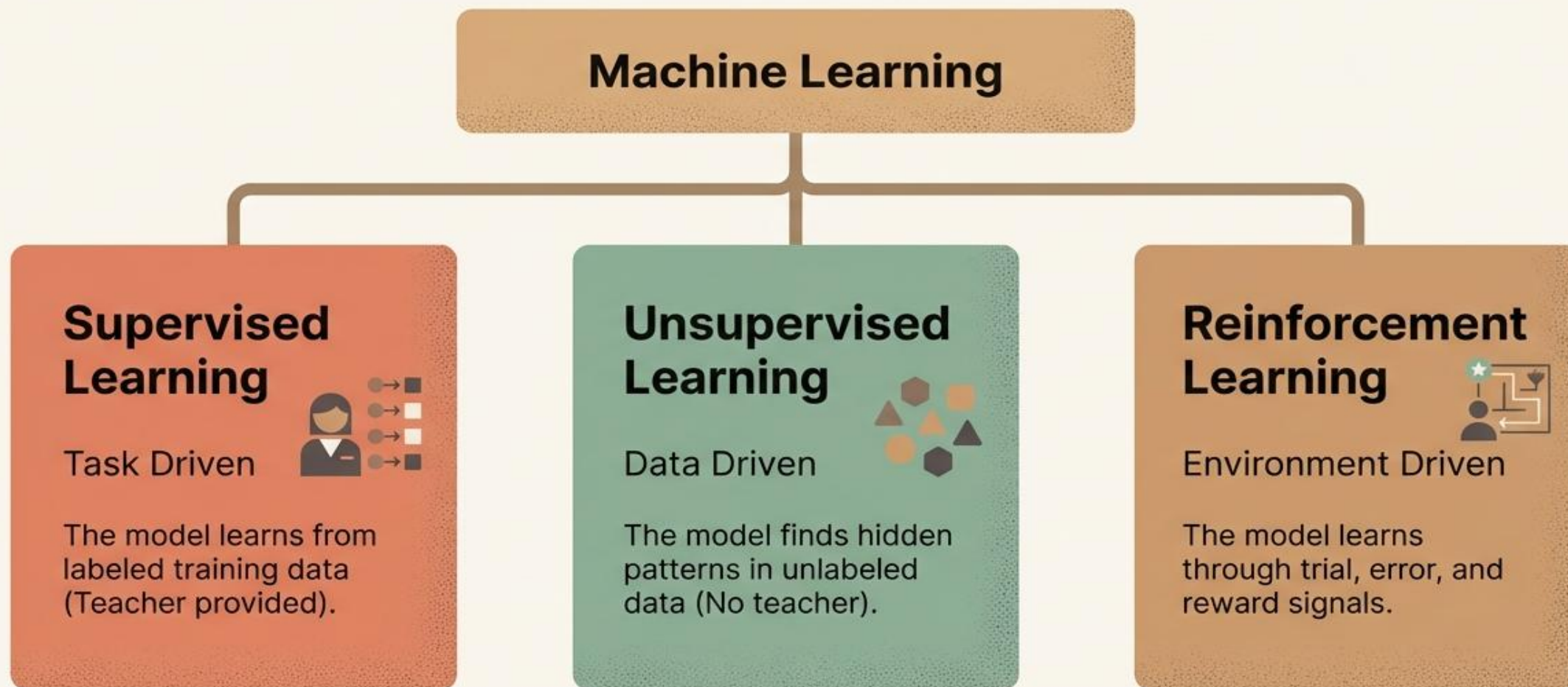


Types of Machine Learning:

Types of
Machine Learning

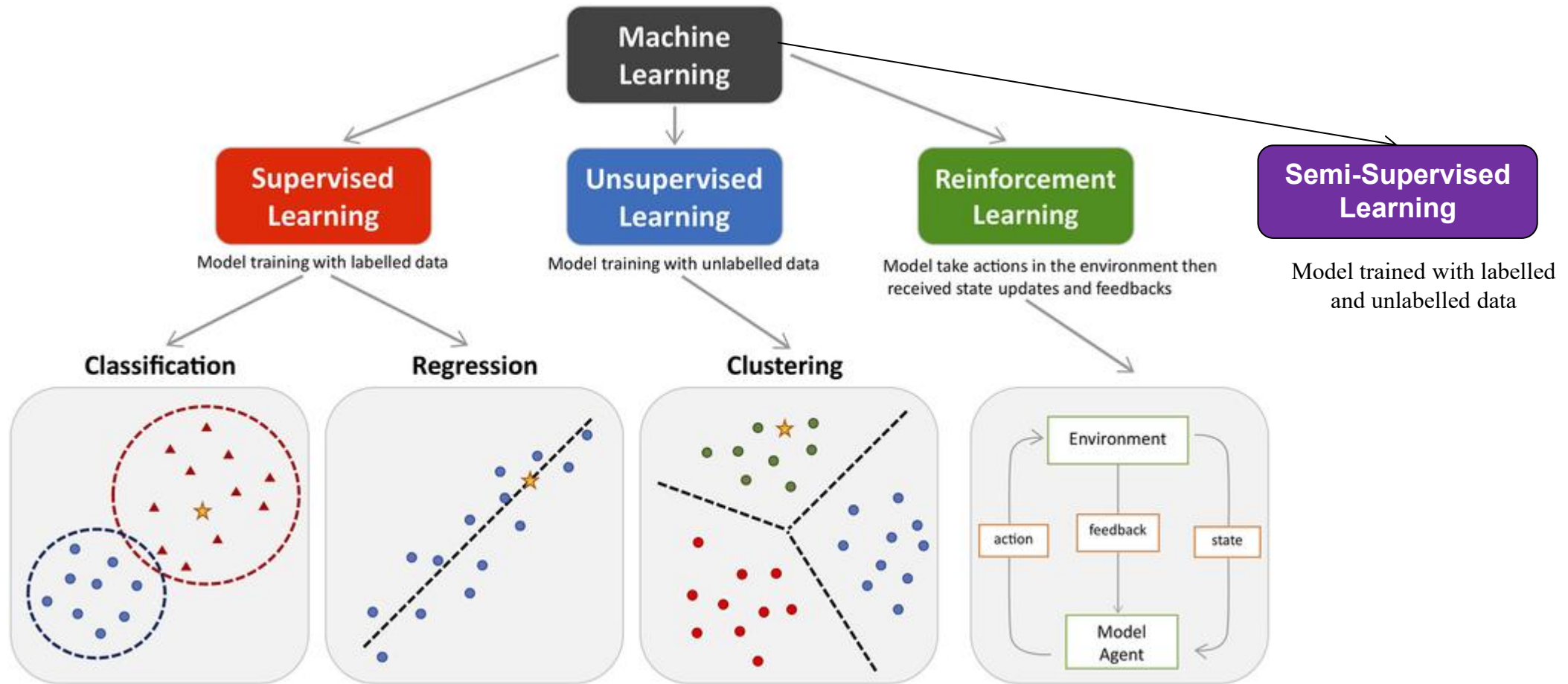
Types of Machine Learning

The Taxonomy of Learning



Note: Semi-supervised learning exists as a hybrid approach.

Types of Machine Learning:

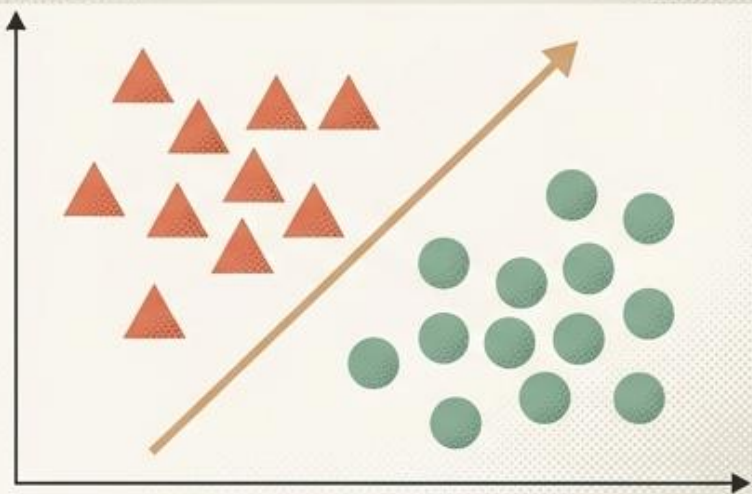


Supervised Learning

Learning with a Teacher

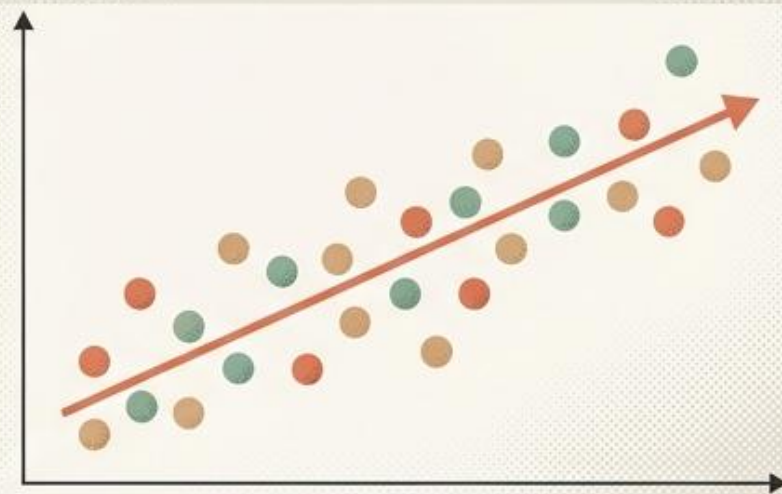
The algorithm is trained on a labeled dataset consisting of inputs (X) and correct outputs (Y). The goal is to learn a mapping function to predict Y for new X.

Classification



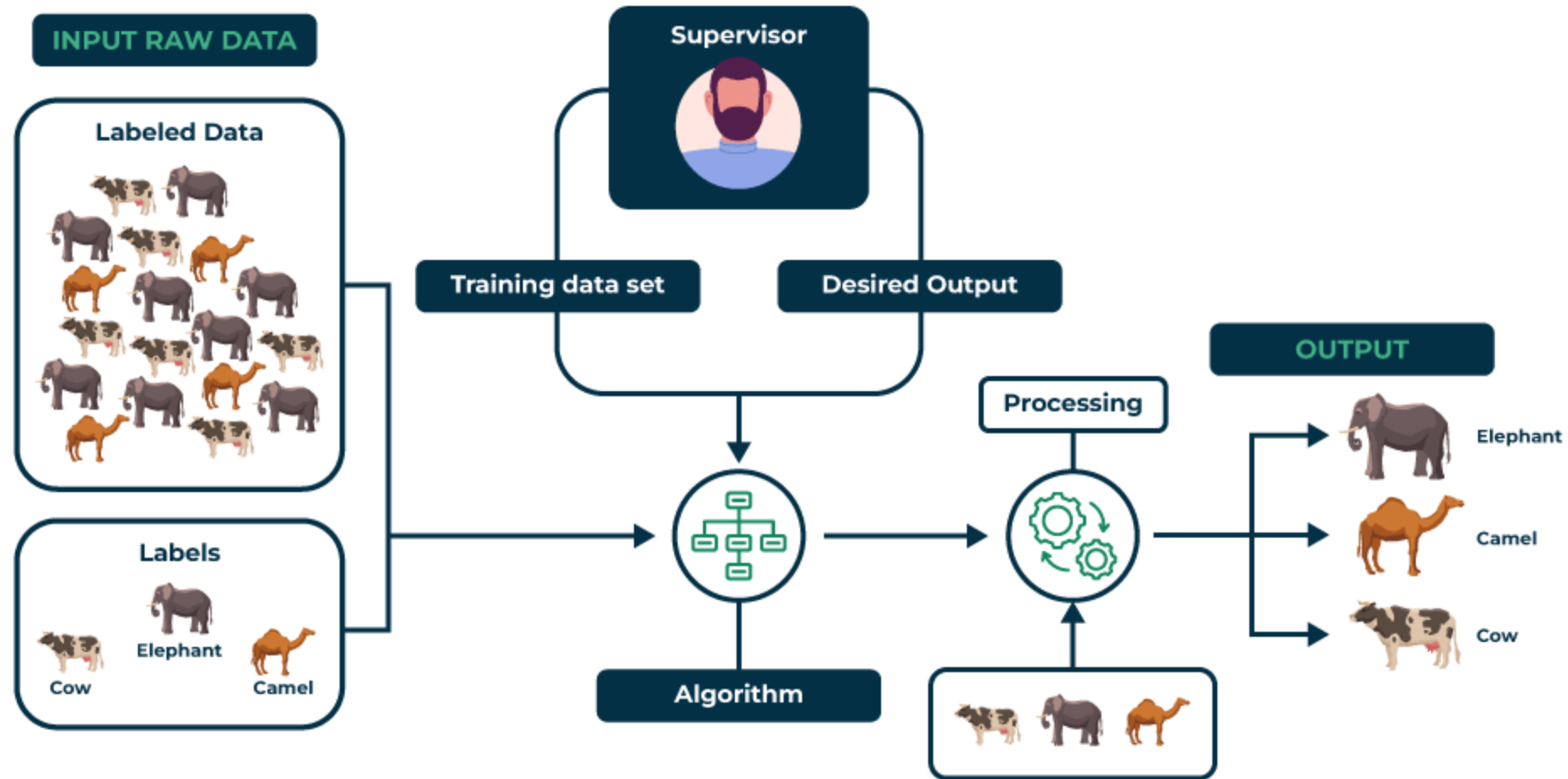
Predicting a Category
(e.g., Spam vs. Not Spam)

Regression



Predicting a Continuous Value
(e.g., House Prices)

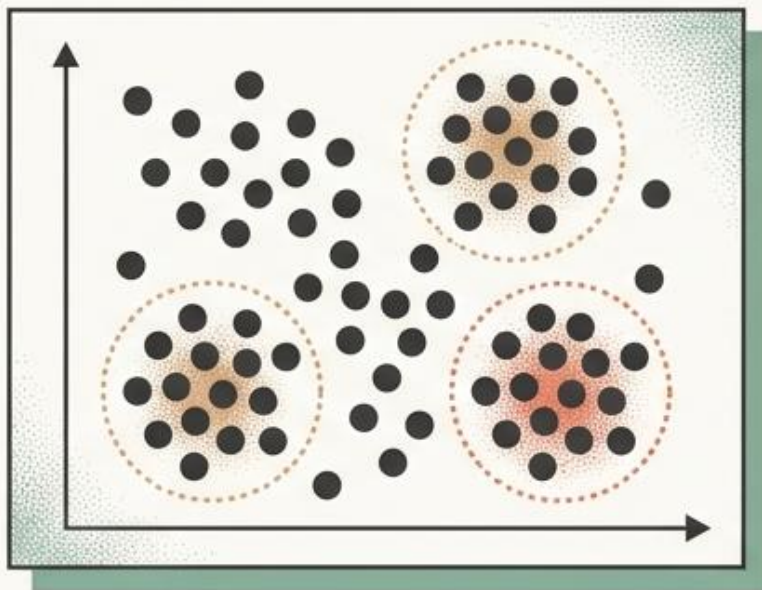
Supervised Learning



Unsupervised Learning

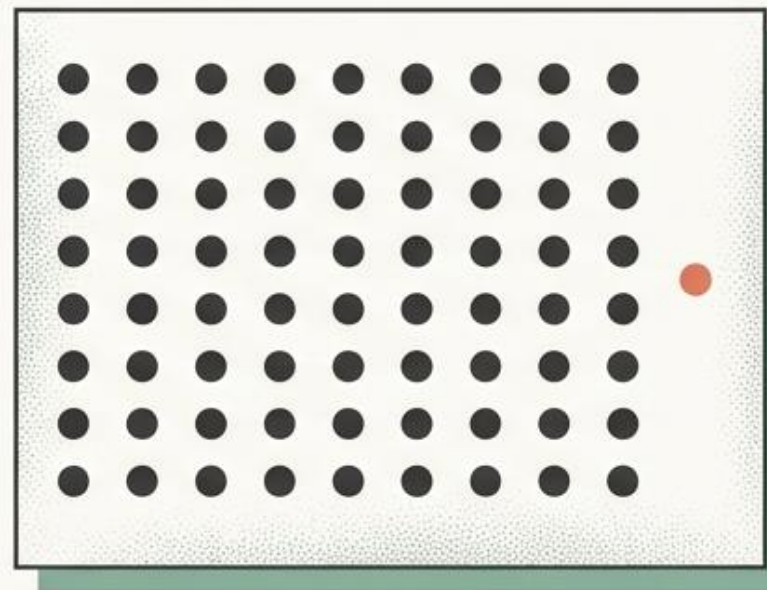
Finding Hidden Structures

The algorithm is given data without explicit instructions or labels. It must explore the data to find inherent structures.



Clustering

Grouping similar data points (e.g., Customer Segmentation)



Anomaly Detection

Identifying outliers (e.g., Fraud Detection)

Unsupervised Learning

INPUT RAW DATA



Interpretation



- Unknown Output
- No Training Data Set

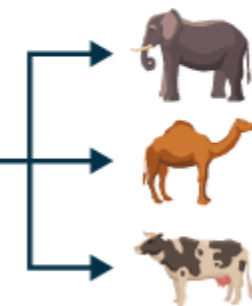
Algorithm



Processing



OUTPUT



MODEL TRAINING

Semi-Supervised Learning

- A machine learning approach that combines supervised and unsupervised learning.
- **Data Used:** Works with both labeled and unlabeled data.
- **Purpose:** Useful when labeled data is costly, time-consuming, or requires specialized skills.
- **Use Case:** Ideal for datasets where a small portion is labeled and the majority is unlabeled.

Technique:

- Apply unsupervised methods to predict labels for unlabeled data.
- Feed these predicted labels into supervised learning algorithms.
- **Common Application:** Frequently used for image datasets, where labeling all images is impractical.

Input Data



Partial Labels



Machine Learning Model



Unlabelled Data



Prediction

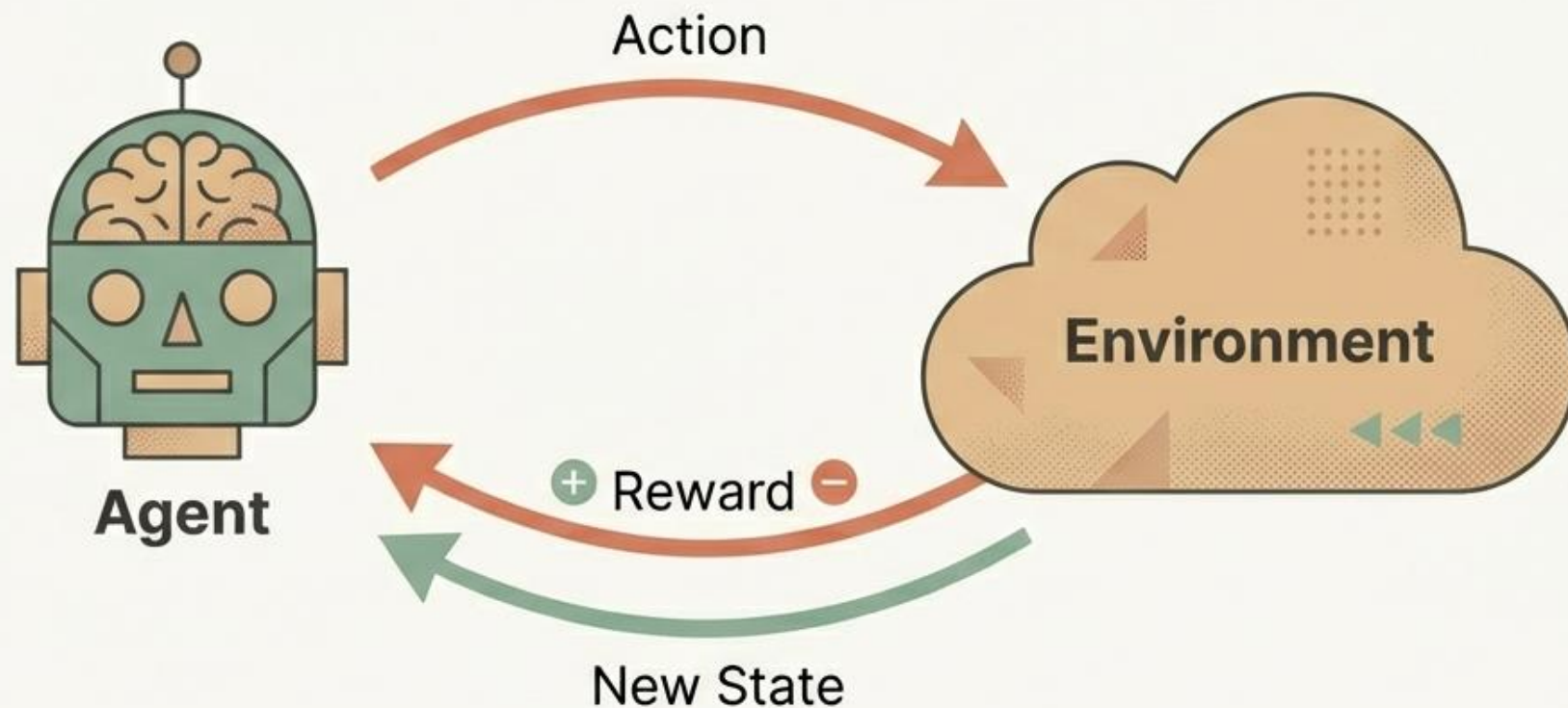
It's an Elephant



Reinforcement Learning

Interaction and Adaptation

An agent learns to make decisions by performing actions in an environment and receiving feedback.



Examples:

- Game Playing (Chess, Go)
- Robotics (Navigation)
- Self-driving Cars (Trajectory Optimization)

- “Reinforcement learning is an autonomous, self-teaching system that essentially learns by trial and error. It performs actions with the aim of maximizing rewards, or in other words, it is learning by doing in order to achieve the best outcomes.”

